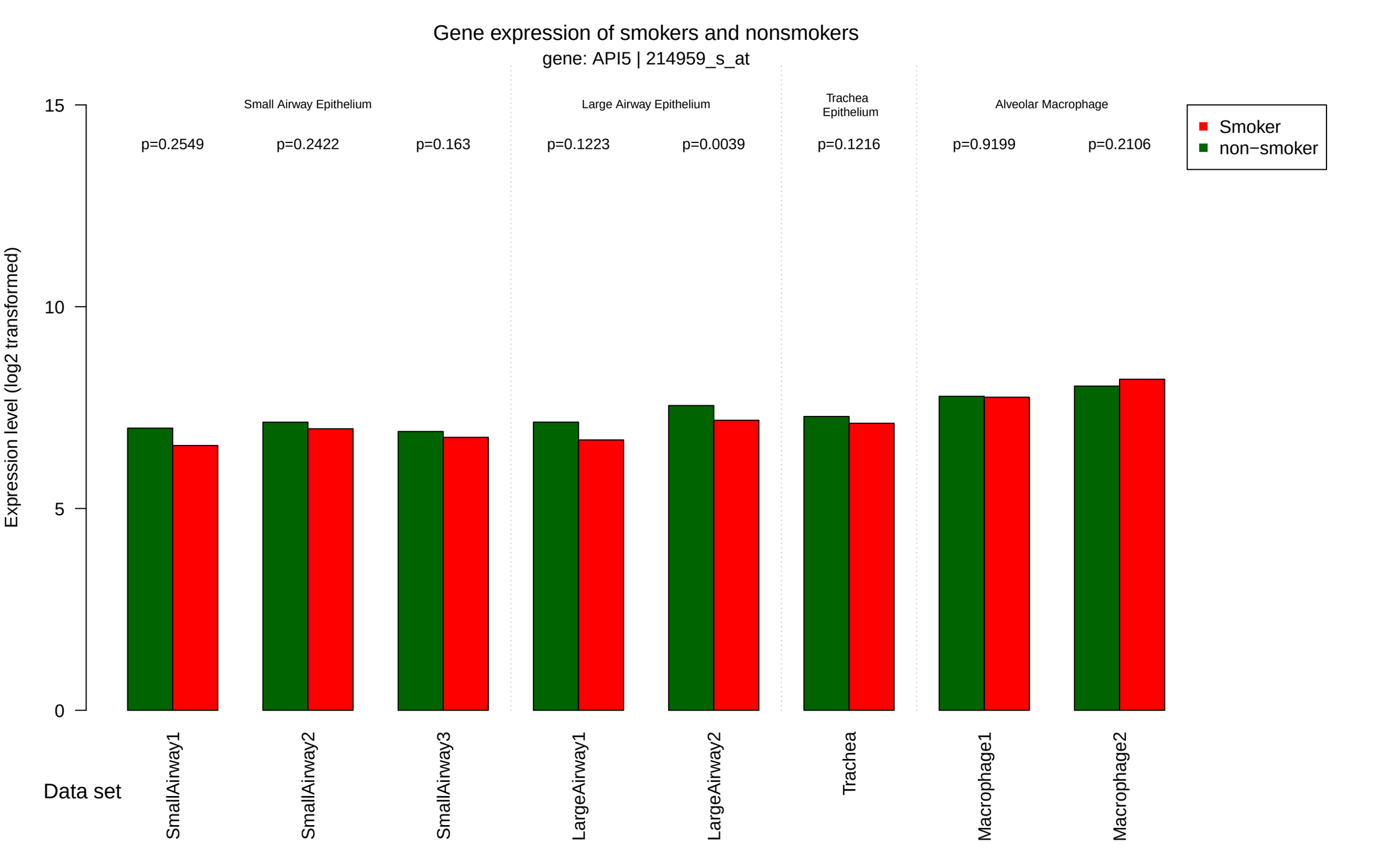
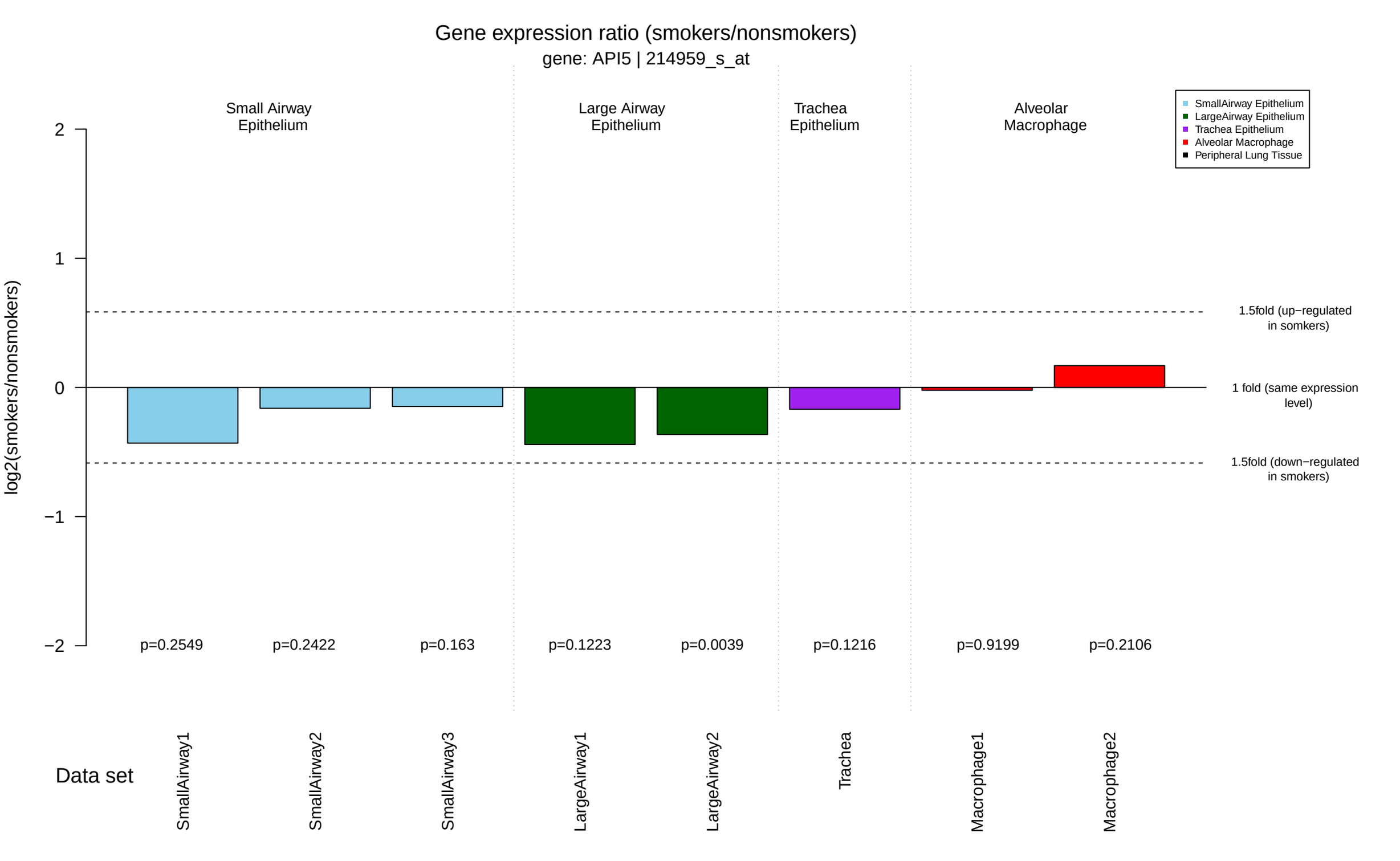
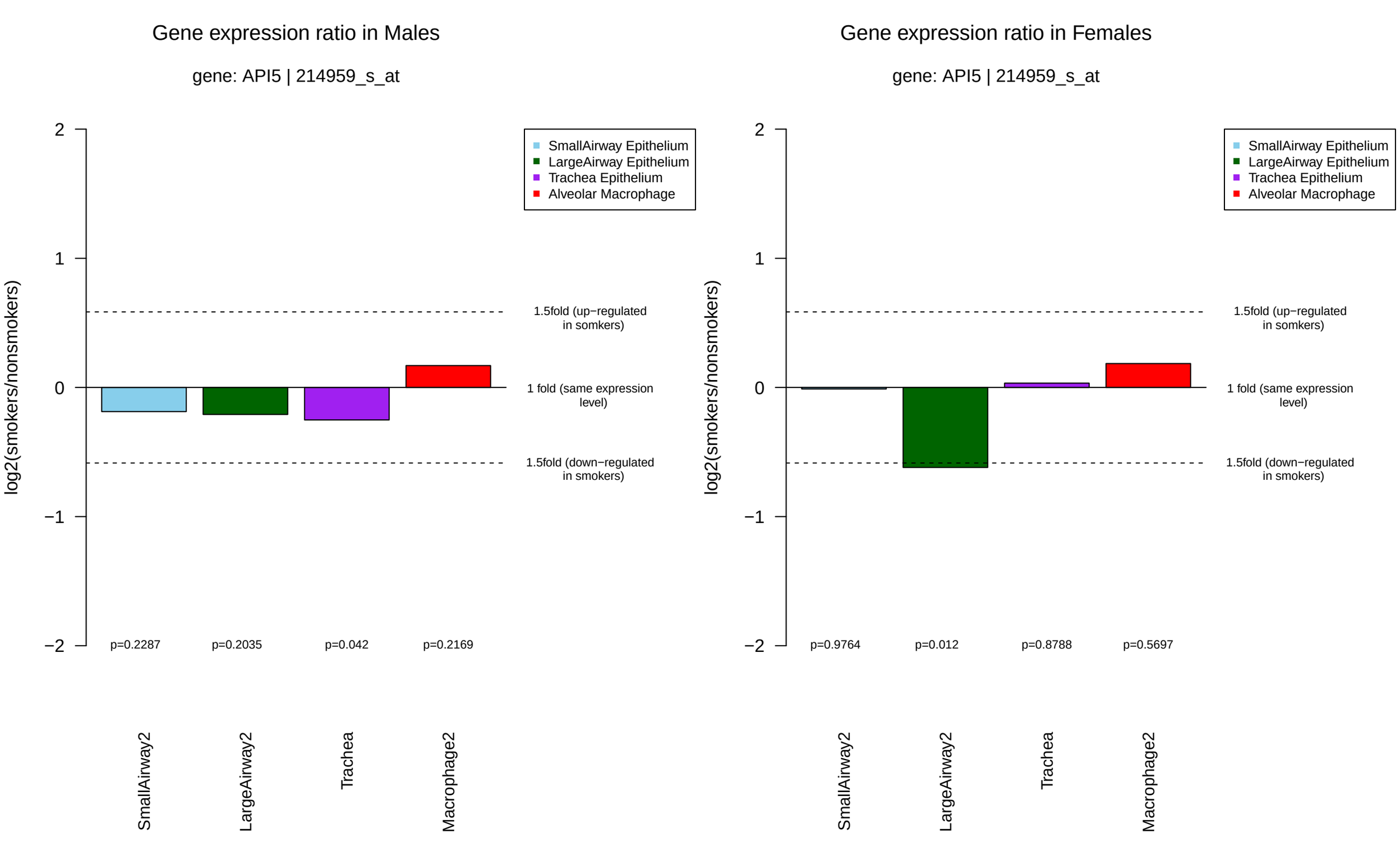
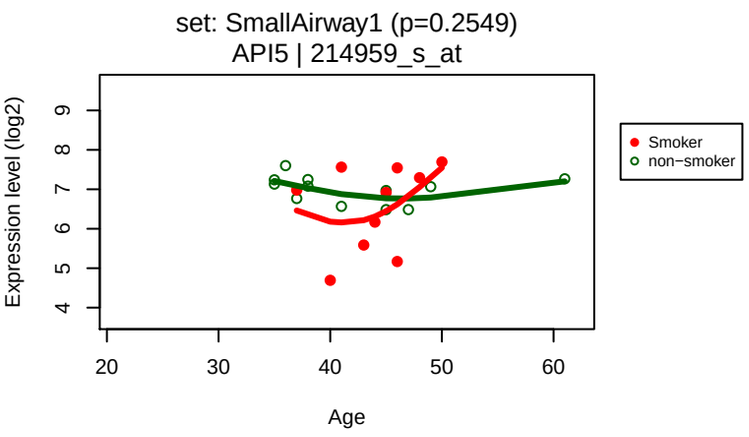
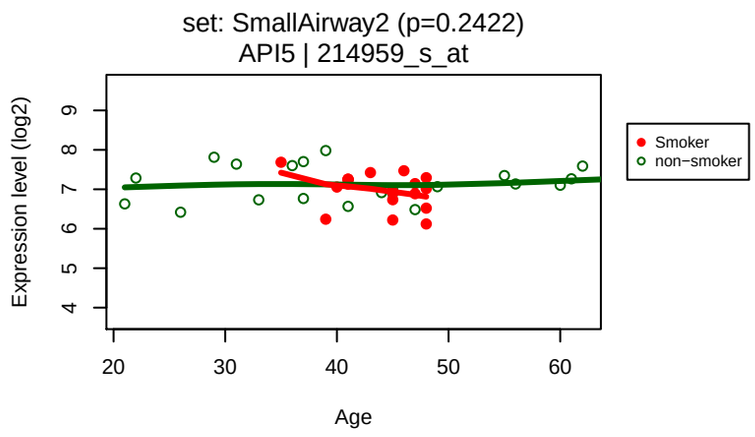
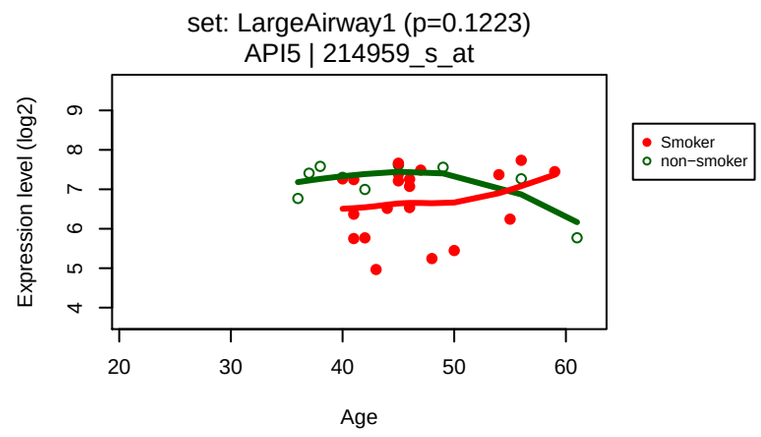

Gene expression of smokers and nonsmokers
gene: API5 | 214959_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2549
p=0.2422
p=0.163
p=0.1223
p=0.0039
p=0.1216
p=0.9199
p=0.2106
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

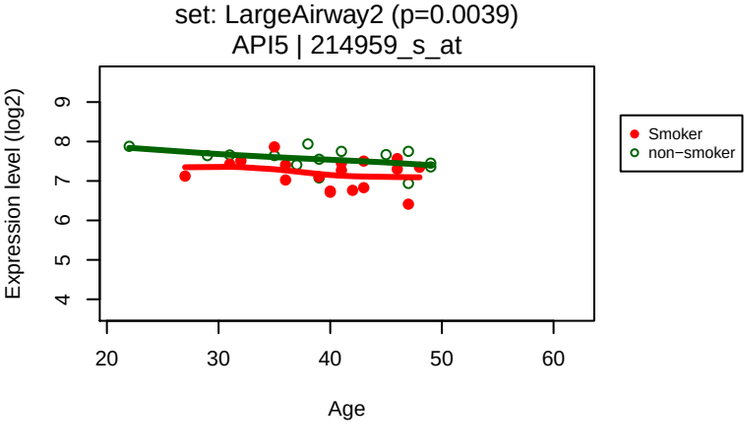

Gene expression ratio (smokers/nonsmokers)
gene: API5 | 214959_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2549
p=0.2422
p=0.163
p=0.1223
p=0.0039
p=0.1216
p=0.9199
p=0.2106
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

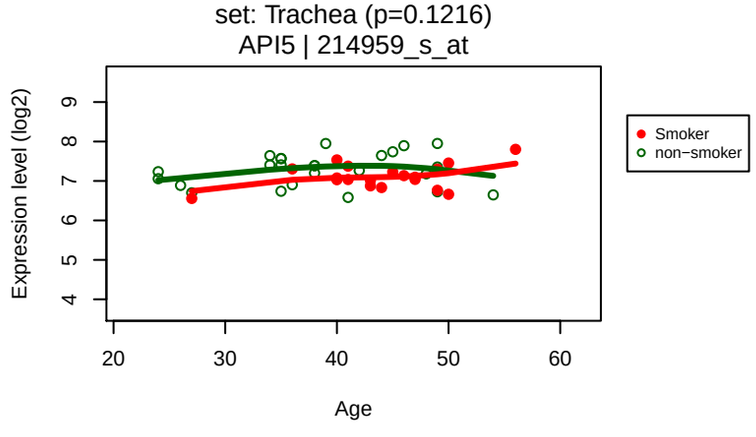

Gene expression ratio in Males
gene: API5 | 214959_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2287
p=0.2035
p=0.042
p=0.2169
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: API5 | 214959_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9764
p=0.012
p=0.8788
p=0.5697
SmallAirway2
LargeAirway2
Trachea
Macrophage2

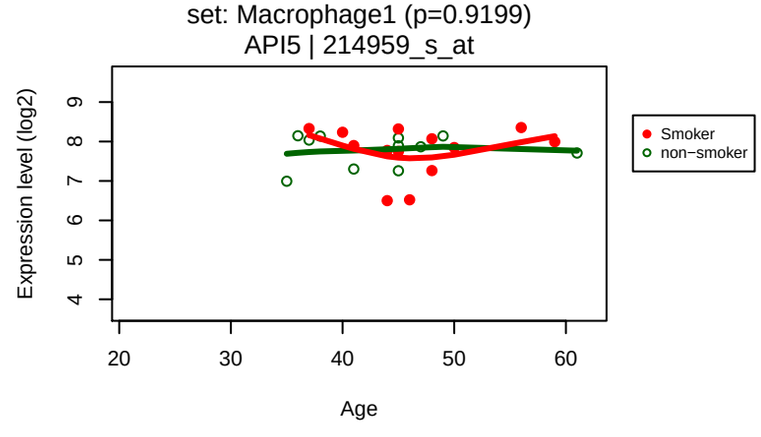

set: SmallAirway1 (p=0.2549)
API5 | 214959_s_at
Expression level (log2)
Age
Smoker
non−smoker

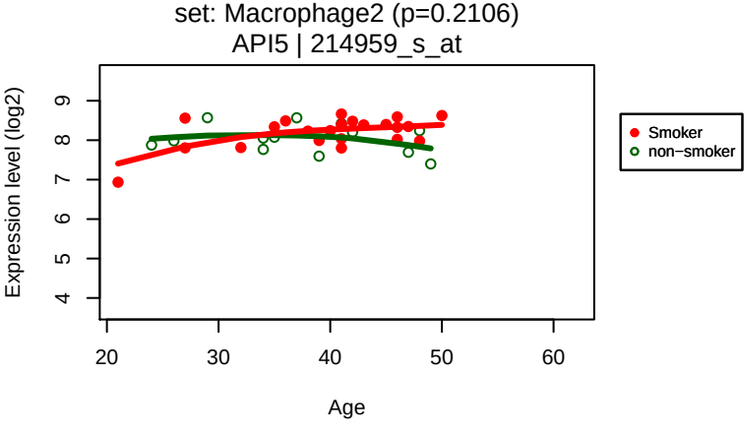

set: SmallAirway2 (p=0.2422)
API5 | 214959_s_at
Expression level (log2)
Age
Smoker
non−smoker

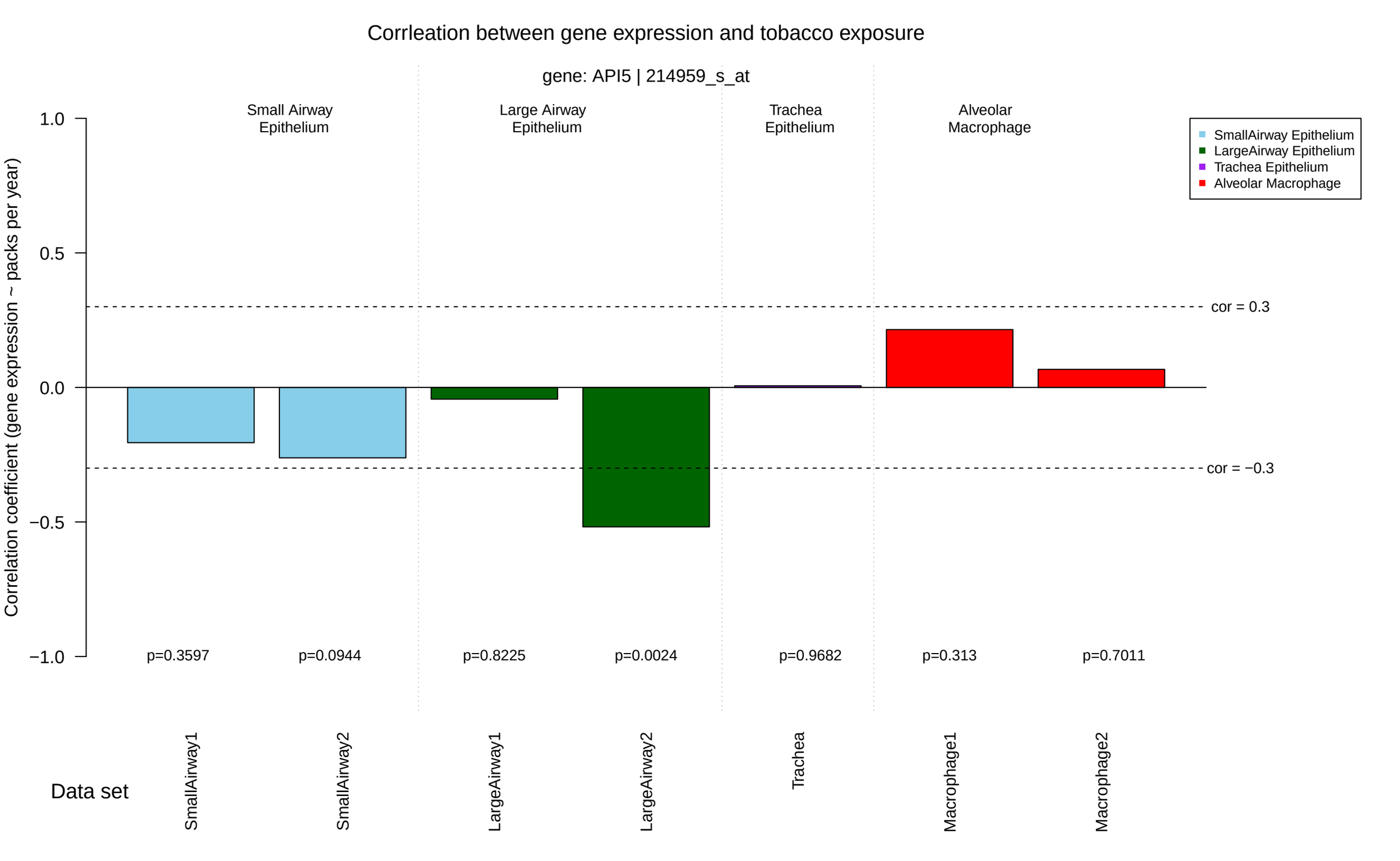

set: LargeAirway1 (p=0.1223)
API5 | 214959_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0039)
API5 | 214959_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1216)
API5 | 214959_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9199)
API5 | 214959_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2106)
API5 | 214959_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: API5 | 214959_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3597
p=0.0944
p=0.8225
p=0.0024
p=0.9682
p=0.313
p=0.7011
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2